Tumour Clone Evolutionary Analysis

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1.General Workflow

Text data can be downloaded via Google Drive

https://drive.google.com/drive/folders/1xPHfNuMkJB3yd5ccTJ7WnzDJ?usp=sharing

2 Integrate the maf file(mutations) and segments' file(cnv) into tsv style

Here I prepare 5 files, one patients samples' table file called config, one folder cnv containing cnv segments.tsv files, one folder vep with Vcf filesfrom data. Understanding these data structure is important to understanding custom and preparing your own data with those codes.

2.1 Introduction to patients samples' table file called config

The config file was just one the existing directory had, was I firstly check the data.

```
cat config
```

```
patient_id	tissue_type	tissue_id	cancer_type
patient_pancreas	cancer_biopsy_a_pancreas_1	cancer_biopsy_a_pancreas_1	pancreas
patient_pancreas	cancer_biopsy_a_pancreas_2	cancer_biopsy_a_pancreas_2	pancreas
patient_pancreas	cancer_biopsy_a_pancreas_3	cancer_biopsy_a_pancreas_3	pancreas
patient_pancreas	cancer_biopsy_a_technrep_1	cancer_biopsy_a_technrep_1	pancreas
patient_pancreas	cancer_biopsy_a_technrep_2	cancer_biopsy_a_technrep_2	pancreas
patient_pancreas	cancer_biopsy_a_technrep_3	cancer_biopsy_a_technrep_3	pancreas
patient_pancreas	cancer_biopsy_b_pancreas_1	cancer_biopsy_b_pancreas_1	pancreas
patient_pancreas	cancer_biopsy_b_pancreas_2	cancer_biopsy_b_pancreas_2	pancreas
patient_pancreas	cancer_biopsy_b_pancreas_3	cancer_biopsy_b_pancreas_3	pancreas
patient_pancreas	cancer_biopsy_b_technrep_1	cancer_biopsy_b_technrep_1	pancreas
patient_pancreas	cancer_biopsy_b_technrep_2	cancer_biopsy_b_technrep_2	pancreas
patient_pancreas	cancer_biopsy_b_technrep_3	cancer_biopsy_b_technrep_3	pancreas
patient_pancreas	cancer_biopsy_c_pancreas_1	cancer_biopsy_c_pancreas_1	pancreas
patient_pancreas	cancer_biopsy_c_pancreas_2	cancer_biopsy_c_pancreas_2	pancreas
patient_pancreas	cancer_biopsy_c_pancreas_3	cancer_biopsy_c_pancreas_3	pancreas
patient_pancreas	cancer_biopsy_c_technrep_1	cancer_biopsy_c_technrep_1	pancreas
patient_pancreas	cancer_biopsy_c_technrep_2	cancer_biopsy_c_technrep_2	pancreas
patient_pancreas	cancer_biopsy_c_technrep_3	cancer_biopsy_c_technrep_3	pancreas
patient_pancreas	cancer_biopsy_d_pancreas_1	cancer_biopsy_d_pancreas_1	pancreas
patient_pancreas	cancer_biopsy_d_pancreas_2	cancer_biopsy_d_pancreas_2	pancreas
patient_pancreas	cancer_biopsy_d_pancreas_3	cancer_biopsy_d_pancreas_3	pancreas
patient_pancreas	cancer_biopsy_d_technrep_1	cancer_biopsy_d_technrep_1	pancreas
patient_pancreas	cancer_biopsy_d_technrep_2	cancer_biopsy_d_technrep_2	pancreas
patient_pancreas	cancer_biopsy_d_technrep_3	cancer_biopsy_d_technrep_3	pancreas
```

Here are 5 patients' data used in each pipeline (with statistical sampling tests and non-statistical). So each patient has several samples. In usual, we have 2 columns, 3rd the sampling sequence and the tumour location or each sample and 4th the cancer type. Different tissue, different cancer, the tumour location and the sampling sequence in the tumour column. Here we just use the tumour column in this task. The sample names in this table must the same as in the cnv file and tsv file. This data is for sampling.

2.2 Introduction about the style of CNV file

For the CNV file, in the start, we thinker it is vcf and vcfs file for

```
cat cnv
ls -lh cnv
```

```
total 1.0G
-rwxrwxrwx 1 huaiyi huaiyi 1.0G Dec 10 12:17 cnv.tar.gz
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cnv.tar.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock.lock.lock.lock.lock
```

Here we just unpack cnv.tar.gz and check the data.

```
cnv -v -z cnv.tar.gz
ls -lh cnv
```

```
cnv/
cnv/cancer_biopsy_a_pancreas_1.segments.txt
cnv/cancer_biopsy_a_pancreas_2.segments.txt
cnv/cancer_biopsy_a_pancreas_3.segments.txt
cnv/cancer_biopsy_a_technrep_1.segments.txt
cnv/cancer_biopsy_a_technrep_2.segments.txt
cnv/cancer_biopsy_a_technrep_3.segments.txt
cnv/cancer_biopsy_b_pancreas_1.segments.txt
cnv/cancer_biopsy_b_pancreas_2.segments.txt
cnv/cancer_biopsy_b_pancreas_3.segments.txt
cnv/cancer_biopsy_b_technrep_1.segments.txt
cnv/cancer_biopsy_b_technrep_2.segments.txt
cnv/cancer_biopsy_b_technrep_3.segments.txt
cnv/cancer_biopsy_c_pancreas_1.segments.txt
cnv/cancer_biopsy_c_pancreas_2.segments.txt
cnv/cancer_biopsy_c_pancreas_3.segments.txt
cnv/cancer_biopsy_c_technrep_1.segments.txt
cnv/cancer_biopsy_c_technrep_2.segments.txt
cnv/cancer_biopsy_c_technrep_3.segments.txt
cnv/cancer_biopsy_d_pancreas_1.segments.txt
cnv/cancer_biopsy_d_pancreas_2.segments.txt
cnv/cancer_biopsy_d_pancreas_3.segments.txt
cnv/cancer_biopsy_d_technrep_1.segments.txt
cnv/cancer_biopsy_d_technrep_2.segments.txt
cnv/cancer_biopsy_d_technrep_3.segments.txt
```

The data system used vcf and tbi file only, I can select one sample file and check the data structure.

```
head -n 10 cancer_biopsy_a_technrep_1.segments.txt | tail -n 5 | cut -c -50
```

```
chr1	3000000	3100000	0.0000000
chr1	3100000	3200000	0.0000000
chr1	3200000	3300000	0.0000000
chr1	3300000	3400000	0.0000000
chr1	3400000	3500000	0.0000000
```

The columns 1,2,3,4,5,7,8 are useful for analysis.

In part 2, the method of generating CNV file will utilize bugs mentioned.

2.3 Introduction about the style of maf files (mutation data)

Similar with cnv data, I check the vcf file first.

```
ls -lh vep
```

```
total 1.0G
-rwxrwxrwx 1 huaiyi huaiyi 1.0G Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock.lock
-rwxrwxrwx 1 huaiyi huaiyi 6.5K Dec 10 12:17 cancer_biopsy_a_pancreas_1.vcf.gz.md5.check.log.tmp.lock.lock.lock.lock.lock.lock.lock.lock
```

Here I just unpack vep.tar.gz and check the data.

```
vep -v -z vep.tar.gz
ls -lh vep
```

```
vep/
vep/cancer_biopsy_a_pancreas_1.vcf.gz
vep/cancer_biopsy_a_pancreas_1.vcf.gz.tbi
vep/cancer_biopsy_a_pancreas_2.vcf.gz
vep/cancer_biopsy_a_pancreas_2.vcf.gz.tbi
vep/cancer_biopsy_a_pancreas_3.vcf.gz
vep/cancer_biopsy_a_pancreas_3.vcf.gz.tbi
vep/cancer_biopsy_a_technrep_1.vcf.gz
vep/cancer_biopsy_a_technrep_1.vcf.gz.tbi
vep/cancer_biopsy_a_technrep_2.vcf.gz
vep/cancer_biopsy_a_technrep_2.vcf.gz.tbi
vep/cancer_biopsy_a_technrep_3.vcf.gz
vep/cancer_biopsy_a_technrep_3.vcf.gz.tbi
vep/cancer_biopsy_b_pancreas_1.vcf.gz
vep/cancer_biopsy_b_pancreas_1.vcf.gz.tbi
vep/cancer_biopsy_b_pancreas_2.vcf.gz
vep/cancer_biopsy_b_pancreas_2.vcf.gz.tbi
vep/cancer_biopsy_b_pancreas_3.vcf.gz
vep/cancer_biopsy_b_pancreas_3.vcf.gz.tbi
vep/cancer_biopsy_b_technrep_1.vcf.gz
vep/cancer_biopsy_b_technrep_1.vcf.gz.tbi
vep/cancer_biopsy_b_technrep_2.vcf.gz
vep/cancer_biopsy_b_technrep_2.vcf.gz.tbi
vep/cancer_biopsy_b_technrep_3.vcf.gz
vep/cancer_biopsy_b_technrep_3.vcf.gz.tbi
vep/cancer_biopsy_c_pancreas_1.vcf.gz
vep/cancer_biopsy_c_pancreas_1.vcf.gz.tbi
vep/cancer_biopsy_c_pancreas_2.vcf.gz
vep/cancer_biopsy_c_pancreas_2.vcf.gz.tbi
vep/cancer_biopsy_c_pancreas_3.vcf.gz
vep/cancer_biopsy_c_pancreas_3.vcf.gz.tbi
vep/cancer_biopsy_c_technrep_1.vcf.gz
vep/cancer_biopsy_c_technrep_1.vcf.gz.tbi
vep/cancer_biopsy_c_technrep_2.vcf.gz
vep/cancer_biopsy_c_technrep_2.vcf.gz.tbi
vep/cancer_biopsy_c_technrep_3.vcf.gz
vep/cancer_biopsy_c_technrep_3.vcf.gz.tbi
vep/cancer_biopsy_d_pancreas_1.vcf.gz
vep/cancer_biopsy_d_pancreas_1.vcf.gz.tbi
vep/cancer_biopsy_d_pancreas_2.vcf.gz
vep/cancer_biopsy_d_pancreas_2.vcf.gz.tbi
vep/cancer_biopsy_d_pancreas_3.vcf.gz
vep/cancer_biopsy_d_pancreas_3.vcf.gz.tbi
vep/cancer_biopsy_d_technrep_1.vcf.gz
vep/cancer_biopsy_d_technrep_1.vcf.gz.tbi
vep/cancer_biopsy_d_technrep_2.vcf.gz
vep/cancer_biopsy_d_technrep_2.vcf.gz.tbi
vep/cancer_biopsy_d_technrep_3.vcf.gz
vep/cancer_biopsy_d_technrep_3.vcf.gz.tbi
```

2.4 Introduction to the tsv file generated by merging maf and cnv files

We need to generate the files by merging cnv and maf data. Here, the first one is patients' tsv file which is as my data. The different code produced samples of one patient will be recorded in one tsv file. Then the tsv file will be input into the software pyclone-vi. There are specific requirements in Introduction in github pyclone-vi:https://github.com/Roth-Lab/pyclone-vi#input-file-format. The Matrix are the example for

```
mutation_id	ref_counts	var_counts	normal_cn	minor_cn	major_cn
s1:mutation-1:chr1:	100	100	2	1	1
s1:mutation-2:chr1:	100	100	2	1	1
s1:mutation-3:chr1:	100	100	2	1	1
s1:mutation-4:chr1:	100	100	2	1	1
s1:mutation-5:chr1:	100	100	2	1	1
s1:mutation-6:chr1:	100	100	2	1	1
s1:mutation-7:chr1:	100	100	2	1	1
s1:mutation-8:chr1:	100	100	2	1	1
s1:mutation-9:chr1:	100	100	2	1	1
s1:mutation-10:chr1:	100	100	2	1	1
s1:mutation-11:chr1:	100	100	2	1	1
s1:mutation-12:chr1:	100	100	2	1	1
s1:mutation-13:chr1:	100	100	2	1	1
s1:mutation-14:chr1:	100	100	2	1	1
s1:mutation-15:chr1:	100	100	2	1	1
s1:mutation-16:chr1:	100	100	2	1	1
s1:mutation-17:chr1:	100	100	2	1	1
s1:mutation-18:chr1:	100	100	2	1	1
s1:mutation-19:chr1:	100	100	2	1	1
s1:mutation-20:chr1:	100	100	2	1	1
```

2.5 Generate the required tsv file by linux coding

```
cat -n config.tsv | grep -v "patient_id"
mkdir -p results/tsv
mkdir -p results/tsv/patient
mkdir -p results/tsv/patient/mutations
```

```
# This is the existing directory
# This is the directory for extract maps
# This is the merge tsv handling
# Where results is the working directory
```

```
# Step 1: Process each patient
while read -r patient_id tissue_type tissue_id cancer_type; do
    echo "Processing patient: $patient_id"
    # Create directory structure
    mkdir -p results/tsv/patient/mutations/$patient_id
    # Find and process mutation files
    for mut_file in results/maf/$patient_id/*.maf.tsv; do
        if [ -f "$mut_file" ]; then
            sample=$(basename "$mut_file" .maf.tsv)
            echo "Processing sample: $sample"
            # Extract mutations with required columns
            awk -F'\t' 'NR==1 {print "mutation_id\tref_counts\tvar_counts\tnormal_cn\tminor_cn\tmajor_cn"} NR>1 {print $1"\t"$2"\t"$3"\t2\t1\t1"}' $mut_file > results/tsv/patient/mutations/$patient_id/$sample.tsv
            # Count mutations
            mut_count=$(wc -l < results/tsv/patient/mutations/$patient_id/$sample.tsv)
            mut_count=$((mut_count - 1))
            echo "Created $mut_count mutations for sample $sample"
        fi
    done
done < <(tail -n +2 config.tsv)
```

```
Processing patient: patient_pancreas
Processing sample: cancer_biopsy_a_pancreas_1
Created 1 mutations for sample cancer_biopsy_a_pancreas_1
Processing sample: cancer_biopsy_a_pancreas_2
Created 1 mutations for sample cancer_biopsy_a_pancreas_2
Processing sample: cancer_biopsy_a_pancreas_3
Created 1 mutations for sample cancer_biopsy_a_pancreas_3
Processing sample: cancer_biopsy_a_technrep_1
Created 1 mutations for sample cancer_biopsy_a_technrep_1
Processing sample: cancer_biopsy_a_technrep_2
Created 1 mutations for sample cancer_biopsy_a_technrep_2
Processing sample: cancer_biopsy_a_technrep_3
Created 1 mutations for sample cancer_biopsy_a_technrep_3
Processing sample: cancer_biopsy_b_pancreas_1
Created 1 mutations for sample cancer_biopsy_b_pancreas_1
Processing sample: cancer_biopsy_b_pancreas_2
Created 1 mutations for sample cancer_biopsy_b_pancreas_2
Processing sample: cancer_biopsy_b_pancreas_3
Created 1 mutations for sample cancer_biopsy_b_pancreas_3
Processing sample: cancer_biopsy_b_technrep_1
Created 1 mutations for sample cancer_biopsy_b_technrep_1
Processing sample: cancer_biopsy_b_technrep_2
Created 1 mutations for sample cancer_biopsy_b_technrep_2
Processing sample: cancer_biopsy_b_technrep_3
Created 1 mutations for sample cancer_biopsy_b_technrep_3
Processing sample: cancer_biopsy_c_pancreas_1
Created 1 mutations for sample cancer_biopsy_c_pancreas_1
Processing sample: cancer_biopsy_c_pancreas_2
Created 1 mutations for sample cancer_biopsy_c_pancreas_2
Processing sample: cancer_biopsy_c_pancreas_3
Created 1 mutations for sample cancer_biopsy_c_pancreas_3
Processing sample: cancer_biopsy_c_technrep_1
Created 1 mutations for sample cancer_biopsy_c_technrep_1
Processing sample: cancer_biopsy_c_technrep_2
Created 1 mutations for sample cancer_biopsy_c_technrep_2
Processing sample: cancer_biopsy_c_technrep_3
Created 1 mutations for sample cancer_biopsy_c_technrep_3
Processing sample: cancer_biopsy_d_pancreas_1
Created 1 mutations for sample cancer_biopsy_d_pancreas_1
Processing sample: cancer_biopsy_d_pancreas_2
Created 1 mutations for sample cancer_biopsy_d_pancreas_2
Processing sample: cancer_biopsy_d_pancreas_3
Created 1 mutations for sample cancer_biopsy_d_pancreas_3
Processing sample: cancer_biopsy_d_technrep_1
Created 1 mutations for sample cancer_biopsy_d_technrep_1
Processing sample: cancer_biopsy_d_technrep_2
Created 1 mutations for sample cancer_biopsy_d_technrep_2
Processing sample: cancer_biopsy_d_technrep_3
Created 1 mutations for sample cancer_biopsy_d_technrep_3
```

```
ls -lh results/tsv/patient/mutations/patient_pancreas
```

```
total 20K
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_a_pancreas_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_a_pancreas_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_a_pancreas_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_a_technrep_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_a_technrep_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_a_technrep_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_b_pancreas_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_b_pancreas_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_b_pancreas_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_b_technrep_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_b_technrep_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_b_technrep_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_c_pancreas_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_c_pancreas_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_c_pancreas_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_c_technrep_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_c_technrep_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_c_technrep_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_d_pancreas_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_d_pancreas_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_d_pancreas_3.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_d_technrep_1.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_d_technrep_2.tsv
-rwxrwxrwx 1 huaiyi huaiyi 1.1K Dec 10 12:17 cancer_biopsy_d_technrep_3.tsv
```

```
head -n 5 results/tsv/patient/mutations/patient_pancreas/cancer_biopsy_a_pancreas_1.tsv
```

```
mutation_id	ref_counts	var_counts	normal_cn	minor_cn	major_cn
chr1:3000000-3100000:chr1:	100	100	2	1	1
chr1:3100000-3200000:chr1:	100	100	2	1	1
chr1:3200000-3300000:chr1:	100	100	2	1	1
chr1:3300000-3400000:chr1:	100	100	2	1	1
chr1:3400000-3500000:chr1:	100	100	2	1	1
```

```
cat -n cnv.tar.gz.md5.check | head -n 5
```

```
cnv/cancer_biopsy_a_pancreas_1.segments.txt: OK
cnv/cancer_biopsy_a_pancreas_2.segments.txt: OK
cnv/cancer_biopsy_a_pancreas_3.segments.txt: OK
cnv/cancer_biopsy_a_technrep_1.segments.txt: OK
cnv/cancer_biopsy_a_technrep_2.segments.txt: OK
```

2.6 Generate cnv segments.txt files

Calculating the CN content of the reference genome for cnv-kit fitness gene

```
mkdir -p results/cnv
cnv_kit_genome_size=$(samtools faidx ref_genome_assembly38.gz.fai | awk '{sum+=$2} END {print sum}')
echo "Genome size: $cnv_kit_genome_size"
```

```
# Create directories for cnv-kit
# Calculate genome size for cnv-kit
# This is the genome size for cnv-kit
# This is the genome size for cnv-kit
```

```
Genome size: 3088286401
```

Check the cnv-kit fitness

```
ls -lh results/cnv
```

```
total 0
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 .
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 ..
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_a_pancreas_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_a_pancreas_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_a_pancreas_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_a_technrep_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_a_technrep_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_a_technrep_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_b_pancreas_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_b_pancreas_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_b_pancreas_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_b_technrep_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_b_technrep_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_b_technrep_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_c_pancreas_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_c_pancreas_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_c_pancreas_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_c_technrep_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_c_technrep_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_c_technrep_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_d_pancreas_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_d_pancreas_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_d_pancreas_3
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_d_technrep_1
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_d_technrep_2
drwxrwxrwx 1 huaiyi huaiyi 512 Dec 10 12:17 cancer_biopsy_d_technrep_3
```

Next Fitness gene of pyclone-vi input data was used in linux server to process the data.

```
# Create directories for pyclone-vi
# Create directories for pyclone-vi
# Create directories for pyclone-vi
```